

Successful desensitization to immunoglobulin A in a case of transfusion-related anaphylaxis

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BACKGROUND: A history of anaphylaxis after transfusion of immunoglobulin A (IgA)-containing blood products in selective IgA-deficient (sIgAD) patients can be a major problem, particularly in emergencies, when large quantities of blood products are required.

CASE REPORT: A 19-year-old woman with end-stage Type 2 autoimmune hepatitis required liver transplantation as her only remaining treatment option. However, she also had sIgAD, anti-IgA antibodies, and episodes of anaphylaxis after receiving IgA-containing blood products. Liver transplantation would have been extremely challenging due to the difficulty of obtaining sufficient blood products from suitable IgA-deficient donors. Hence, it became imperative to devise a protocol to desensitize her to IgA-containing blood products.

RESULTS: Using a continuous infusion of an IgA-enriched (6 mg/mL IgA) immunoglobulin preparation with gradual increases in concentration, she was successfully desensitized to IgA. Consequently, she was able to receive standard platelets, fresh-frozen plasma, and red blood cells with no complications.

CONCLUSION: This approach could prove very useful in similar cases that may require administration of large quantities of blood products particularly in emergency lifesaving circumstances.

Anti-immunoglobulin A (IgA) antibodies are found in approximately one-third of patients with selective IgA deficiency (sIgAD) and can be associated with severe transfusion reactions including anaphylaxis,¹ if patients receive blood products containing IgA from normal donors. Difficulties arise when such patients require treatment with large quantities and many different blood products. We present a patient with end-stage autoimmune hepatitis, sIgAD, and IgA-related anaphylaxis. The only remaining treatment option, liver transplantation, would have been a significant undertaking due to the scarcity of suitable IgA-deficient blood donors. There are two reports of successful liver transplantation in IgA-deficient patients with a history of transfusion reactions to IgA-containing blood products.^{2,3} In one of these reports,² the patients did not have a very serious transfusion reaction and tolerated washed red blood cells (RBCs) and standard platelets (PLTs). However, our patient did not tolerate RBCs or PLTs treated with washing or leukoreduction, respectively. In the other report,³ elaborate logistical measures were taken to maintain the availability of viable IgA-deficient PLTs because of the short PLT shelf life of only 5 days. This approach, although possible, would have reduced the chances of receiving a transplant within the limit of time dictated by the terminal nature of our patient's illness. Therefore, in view of the very good outcomes of liver transplantation for autoimmune hepatitis,⁴ we desensitized this patient to IgA to allow the use of blood products so that transplantation could proceed.

ABBREVIATION: sIgAD = selective IgA deficiency.

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CASE REPORT

A 19-year-old who at the age of 11 developed autoimmune hypothyroidism was diagnosed with liver kidney microsomal-positive autoimmune hepatitis at the age of 16 years. During the investigation of her liver disease, she received an infusion of PLTs for thrombocytopenia before a liver biopsy. Approximately 10 minutes into the transfusion, she became short of breath and wheezy, developed central abdominal pain and a widespread rash all over her body, and became hypotensive (blood pressure

96/48 mmHg, pulse 88). She had two further episodes of anaphylactic reactions to RBCs, cryoprecipitate, and PLTs. She was investigated and found to have sIgAD with anti-IgA antibodies detected by passive hemagglutination assay. A diagnosis of IgA-related anaphylaxis was confirmed on the basis of transfusion-associated anaphylaxis after administration of IgA-containing blood products on more than one occasion, IgA deficiency, and the presence of anti-IgA antibodies.

Progression of her liver disease resulted in portal hypertension, esophageal varices, clotting abnormalities, and episodes of gastrointestinal bleeding. She did not tolerate washed standard RBCs or leukoreduced PLTs; therefore, RBCs, fresh-frozen plasma (FFP), and PLTs, from IgA-deficient donors,⁵ were required to treat the episodes of gastrointestinal bleeds.

By 19 years of age, her autoimmune liver disease had progressed to end-stage hepatic failure requiring transplantation. However, significant difficulty was anticipated for the procedure because of the limited availability of suitable blood products from IgA-deficient donors to support liver transplantation. Consequently, placement on the liver transplant waiting list was deferred and therefore a protocol for desensitization to IgA was devised on the basis of a few reported observations and the protocols used for desensitization against protein compounds.^{6,7}

METHODS AND RESULTS

IgA levels were measured using the turbidometry method on a clinical chemistry system (ADVIA 2400, Bayer UK Ltd, Cambridge, UK). In addition, anti-IgA and IgA antibody levels were measured at the Red Cell Immunohematology Laboratory, NHS National Blood Services (Colindale, UK) by the hemagglutination or the hemagglutination inhibition assays using the method of Hunt and colleagues,⁸ respectively. Briefly, blood group O-negative RBCs were coated with IgA1 or IgA2 myeloma proteins and, as controls, RBCs were either coated with random blood group AB serum or were left uncoated. Anti-IgA in test sera was detected by hemagglutination induced by titrations of test sera. IgA in test sera was detected by inhibition of hemagglutination induced by commercially obtained anti-IgA antibodies (Dako UK Ltd, Ely, UK).

Before desensitization, skin prick tests were performed by placing drops of 10-fold dilutions of the test antigen (1:10,000-1:1) on the volar aspect of the forearm followed by breaching the skin barrier through the drops using a sterile lancet as per standard protocol.⁹ The test antigens were Intratect (50 mg/mL, 96% IgG, 2 mg/mL IgA, Biotest, Dreieich, Germany) or Pentaglobin (36 mg/mL IgG, 6 mg/mL IgM, and 6 mg/mL IgA, Biotest). The response was measured after 15 minutes. The intradermal skin tests were done by intradermal injections of 25 μ L of 10-fold dilutions (1:10,000-1:1) of the test antigens in the volar aspect of the forearm and measuring the responses after 15 minutes. Both skin prick and intradermal tests were negative at all dilutions in the presence of a positive reaction to histamine.

One day later the patient was desensitized. At the time of desensitization the patient was on thyroxine, ciprofloxacin, spironolactone, calcium supplements, multivitamins, vitamin K, and ranitidine. She was taking 750 mg of mycophenolate mofetil twice a day and 5 mg of prednisolone once a day for treatment of autoimmune hepatitis. Desensitization was performed by slow infusion (0.4 mL/kg/hr) of serial dilutions of Pentaglobin starting at a dilution of 10^{-7} . Every 30 minutes the concentration was increased 10-fold until a neat solution of Pentaglobin was infused. No side effects were detected; hence an additional 100 mL of Pentaglobin was infused at the same rate (32 mL/hr). The next day, the patient was challenged with the intravenous (IV) infusion of 3 mL of standard PLTs with no adverse effects. This was followed by infusion of 30 mL and finally 300 mL of PLTs with no side effects. She then tolerated 3 units of standard FFP to correct her international normalized ratio for a planned paracentesis. Two days later she received 3 units of standard RBCs with no adverse reactions in the course of emergency treatment of bleeding esophageal varices. To maintain this state of unresponsiveness, weekly IV infusions of IgA-enriched immunoglobulins had been administered. The weekly infusions would continue until after a successful liver transplantation, when the case would be reappraised. This regimen had so far resulted in maintenance of a state of unresponsiveness as anti-IgA antibodies could not be detected either immediately after desensitization or 1 month later (Table 1). In addition, by 3 months postdesensitization the patient had received 6 units of random

TABLE 1. Changes in serum antibodies, complement, and tryptase in relation to desensitization and challenge

Time	IgA (g/L)	Anti-IgA antibodies	Tryptase (μ g/L)	C3 (g/L)	C4 (g/L)
Predesensitization	0.03	POS	1.28	0.95	0.21
1 day postdesensitization	0.72	ND	1.09	0.65	0.09
1 day postchallenge with PLTs and FFP	0.39	ND	1.27	0.63	0.11
27 days postdesensitization	0.36*	ND			

* Trough level of weekly Pentaglobin.

ND = not detectable; POS = positive (titer range, 2-16).

RBCs, 9 units of random PLTs, and 15 units of random FFP without any adverse events.

DISCUSSION

The diagnosis of IgA-related anaphylaxis is made on the basis of IgA and anti-IgA measurements after transfusion-related anaphylaxis. Detection of IgA would exclude a diagnosis of sIgAD and IgA-related anaphylaxis. In sIgAD patients with a history of anaphylaxis, blood components from IgA-deficient donors are used unless in an emergency where such products are not available. Under these circumstances, controlled transfusion of standard leukoreduced RBCs, PLTs, and plasma is recommended.⁵ However, our patient had anaphylactic reactions to PLTs and even to RBCs that had undergone leukoreduction and washing, respectively. Therefore, considering the potential need for large-volume and emergency transfusions for liver transplantation and the added complication of a history of esophageal varices and gastrointestinal bleeding, in our patient, we could not rely on the methods used previously^{2,3} for management of IgA-deficient patients undergoing liver transplantation.

Three conference abstracts have reported methods of giving intravenous immune globulin (IVIG) in a form that was tolerated by patients who previously had reactions to IgA-containing IVIG. One abstract¹⁰ reported that nine patients who had not tolerated immunoglobulin products containing 0.7 g/L of IgA did tolerate "IgA-poor" (IgA < 0.001 g/L im, IgA < 0.004 g/L IV) immunoglobulin preparations. In the second abstract,¹¹ it was shown that a 6-year-old child who had anaphylactic reactions to IVIG, with an unknown anti-IgA status, on one occasion tolerated incremental (140 mL/hr) IV infusion of 5% Gamagard, containing less than 1 g/L of IgA, when premedicated with methylprednisolone and diphenhydramine hydrochloride. In the third,¹² graded subcutaneous immunoglobulin injections were tolerated by an IgA-deficient patient who had a history of anaphylactic reactions to IVIG (brand or IgA content not mentioned). This patient did not have anti-IgA antibodies before the procedure and subsequently tolerated infusions of the actual agent used for desensitization. However, two published observations alerted us to the possibility of desensitization to IgA in our patient. The first interesting observation was an incidental finding of reduction or loss of anti-IgA antibodies in one sIgAD⁷ and three common variable immunodeficiency⁶ patients after subcutaneous infusion of an immunoglobulin preparation containing up to 5 mg/mL IgA. The second observation was a report of two common variable immunodeficiency patients with IgA deficiency and a history of anaphylactic reactions to IgA-containing IVIG. In one patient, slow infusion (10 g in 8 hr) of IVIG was tolerated and was associated with disappearance of circulating anti-IgA antibodies. Subsequent infusions at the usual rate

were also tolerated. In the other patient, autologous plasma was mixed with an IgA-containing immunoglobulin preparation *ex vivo* and then transfused with no side effects. The anti-IgA activity diminished and the patient subsequently received IgA-containing IVIG with no adverse effects.

It is likely that IgA-related anaphylaxis is caused by more than one mechanism as anti-IgA antibodies cannot be demonstrated in all IgA-deficient patients who have posttransfusion anaphylactic reactions. While it might be argued that this suggests that anti-IgA antibodies do not contribute to the mechanism leading to IgA-related anaphylaxis, it has been shown that a significantly higher proportion of IgA-deficient patients with anaphylaxis have anti-IgA antibodies compared to IgA-deficient patients with no history of anaphylaxis. In one study,¹³ 61 of 80 (76.3%) IgA-deficient patients with transfusion-related anaphylaxis had anti-IgA antibodies, whereas only 21 of 97 (21.7%) IgA-deficient patients with no history of anaphylaxis had anti-IgA antibodies. The majority of case reports and studies describe the presence of anti-IgA antibodies of the IgG class,^{6,14} but some have demonstrated IgE anti-IgA antibodies in association with anaphylaxis.^{15,16} This lends further support to the hypothesis that different mechanisms are involved in anti-IgA-related anaphylaxis. Where IgE is involved, anaphylaxis is likely to be a Type 1 hypersensitivity reaction despite the absence of any reports of positive skin tests in these cases. On the other hand, based on murine models, an alternative mechanism of anaphylaxis mediated by IgG and basophils has been proposed.^{17,18} This pathway is associated with an increase in anaphylatoxins C3a and C5a but not in tryptase. An increase in C3a and C4a without an infusion reaction has also been reported after IV administration of immunoglobulins with low IgA (0.4–2.9 µg/mL) in patients with anti-IgA-associated anaphylaxis.¹⁹ In our patient, tryptase levels did not change with desensitization. However, loss of anti-IgA activity was associated with a reduction in the level of complement components C3 and C4 (Table 1). This implies generation of anaphylatoxins (C3a and C5a); however, the kinetics were probably too slow to result in anaphylaxis. In addition, given that we have not detected cutaneous sensitization to IgA and that anti-IgA hemagglutinins of the IgG class have been detected in our patient's plasma, we speculate that the mechanism of anti-IgA-related anaphylaxis in our case is most probably IgG-mediated.

In general, desensitization is mainly performed for common allergens. This is done either by repeated exposures to very small doses of the antigen or by gradual introduction of increasing concentrations of the antigen. The mechanism of desensitization is poorly understood and is likely to differ when desensitizing different immune responses. In an *in vitro* murine model of IgE-mediated anaphylaxis, using a knock-out mouse for signal

transducer and activator of transcription (STAT)-6, it has been shown that the process of desensitization of marrow-derived mast cells is an active process that involves sub-optimal activation of mast cell high-affinity IgE receptor, FCεRI.²⁰ However, in the majority of IgA-related transfusion reactions the anti-IgA antibodies are of IgG subclass. It is possible that introduction of the antigen, in this case IgA, in the context of an immunoglobulin preparation resulted in engagement of inhibitory receptors by IgG and control basophil or mast cell degranulation while anti-IgA was being neutralized. In a mouse system where IgE-mediated anaphylaxis was induced by prior sensitization to a model antigen—interestingly, in this case an immunoglobulin, goat IgG—it has been demonstrated that blocking mouse antibodies prevented anaphylaxis²¹ by at least two mechanisms. The first mechanism was through interception of the antigen by mouse IgG before the antigen could induce target cell (mast cells or basophils) activation and degranulation. The second mechanism was by binding and cross-linking the inhibitory receptors FCγRIIb to the activating receptors FCεRI on target cells.

This case illustrates for the first time the use of IgA-enriched immunoglobulin preparations as a source of antigen for desensitization in cases of IgA-related post-transfusion anaphylaxis. Currently in this patient, as a result of IVIG infusions a significant level of serum IgA (Table 1) is detected without any evidence of an immunologic reaction. Desensitization should be considered in cases with a history of IgA-related anaphylaxis who suffer from other medical conditions requiring frequent administration of blood products as an emergency lifesaving treatment. Considerably more experience is needed to determine the efficacy of this approach and hence such patients should be referred to immunologists to be investigated and considered for desensitization. This approach may be effective in at least two groups of IgA-deficient patients: 1) those with a history of anaphylactic transfusion reactions and anti-IgA antibodies or 2) patients in whom anti-IgA antibodies are not detected but have a definite history of tolerating IgA-deficient products.

CONFLICT OF INTEREST

SKA, PFKY, DGK, SEH, AM, and ARS declare no conflict of interest. MAAL is a member of the medical advisory boards for Baxter Healthcare Ltd and Biotest (UK) Ltd.

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